

Question	Answer	Mark	Mark scheme	Additional guidance
18	Proof	M1	begins proof to show that triangles ABO and CBO or triangles ABD and CBD are congruent by giving one pair of equal sides or equal angles with reason	Where D is point such that BOD is diameter
		M1	for different pair of equal sides or angles with reason	
		C1	for full proof that triangles ABO and CBO are congruent, SSS, or triangles ABD and CBD are congruent, RHS, and therefore angle $ABO = \text{angle } CBO$ $AB = CB$ (given) BO (or BD) is <u>common</u> $AO = CO$ <u>radii</u> of circle angle $BAD = \text{angle } BCD$ <u>angles</u> in a <u>semicircle</u> are 90 ($BO = AO = CO$ <u>radii</u> of circle) counts as two sides with reasons OR	
		M1	draws OA , OC and AC and labels angle $OAC = x$ and angle $OCA = x$ with reason given, $AO = CO$ <u>radii</u> of circle and base angles of an <u>isosceles triangle</u> are equal or $BAC = BCA$ since ABC is isosceles	
		M1	shows $OAC = OCA$ and shows $BAC = BCA$ and uses these to show $OAB = OCB$ with all reasons given	
		C1	for full proof concluding with angle $ABO = y$ and angle $CBO = y$ with reason given, eg $OA = OB = OC$ radii of circle and OBC and OAC are isosceles	

Documented scheme clarification

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Original source

scheme page 17, SHA256

9d7ee2fd705b607c70423ec1b7fedc4940917c4295e50bc0b3d3cb95cc1b1fa8

Original wording

for full proof concluding with angle $ABO = y$ and angle $CBO = y$ with reason given, eg $OA = OB = OC$ radii of circle and OBC and OAC are isosceles

Correction / interpretation

In the final C1 guidance for the alternative proof, replace the triangle reference ' OBC and OAC are isosceles' with ' OBC and OAB are isosceles'.

Reason

$OA = OB$ gives angle $ABO =$ angle OAB in triangle OAB . $OB = OC$ gives angle $CBO =$ angle OCB in triangle OBC . The preceding step establishes angle $OAB =$ angle OCB , so angle $ABO =$ angle CBO . OAC is also isosceles, but OAB is the triangle required for this final inference.

Sol authority

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